Conceptual question · Unit 2 · 2 points · 15 min · Conceptual / fast

CO2 — Motion in a cubic potential: a separatrix that is never crossed and an asymmetric well

A point particle P of mass m moves along the x axis of an inertial frame under the single conservative force derived from the potential energy

$$V(x) = V_0 \left[\left(\frac{x}{b} \right)^3 - 3\,\frac{x}{b} \right],$$

with $V_0 > 0$ (an energy) and $b > 0$ (a length) constants. The figure shows $V(x)$ and two energy levels. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

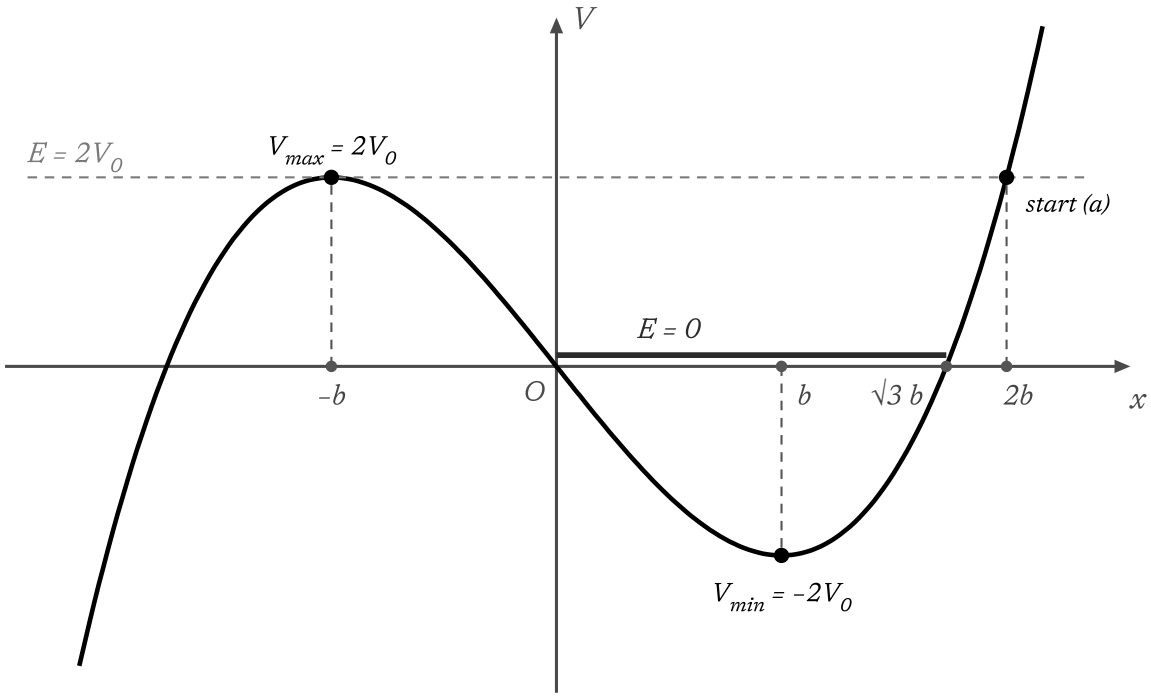


Figure 1. Potential $V(x) = V_0\left[(x/b)^3 - 3x/b\right]$ with the energy levels $E = 2V_0$ (released at $x = 2b$) and $E = 0$ (released at $x = 0$).

Part (a) · 1 point

Released at $x = 2b$

Statement: "If P is released from rest at $x = 2b$, it moves to the left, reaches the top of the barrier at $x = -b$ after a finite time and then escapes towards $x \to -\infty$."

Step 1. Energy and direction of the motion

Write $u = x/b$, so $V = V_0(u^3 - 3u)$. By the chain rule $\frac{dV}{dx} = \frac{dV}{du}\frac{du}{dx} = \frac{3V_0}{b}(u^2 - 1)$: the equilibria are $x = \pm b$, with $V(-b) = 2V_0$ (maximum, $V'' < 0$) and $V(b) = -2V_0$ (minimum). Released from rest at $x = 2b$, the energy is $E = V(2b) = V_0(8 - 6) = 2V_0$ and the initial force is $F = -\frac{dV}{dx}\big|_{2b} = -\frac{3V_0}{b}(4 - 1) = -\frac{9V_0}{b} < 0$: P moves to the left.

Step 2. The kinetic energy factorises

Energy conservation (Notes §4.5) gives $\frac{1}{2}m\dot{x}^2 = E - V$, and

$$E - V = V_0\big(2 - u^3 + 3u\big) = V_0\,(u+1)^2\,(2-u)$$

(check: $(u+1)^2(2-u) = (u^2 + 2u + 1)(2 - u) = -u^3 + 3u + 2$). For $-1 < u < 2$ it is positive, and it vanishes at $u = -1$ with a **double** root.

Step 3. Time to approach $x = -b$

On the way to the left $\dot{x} = -\sqrt{2V_0/m}\,(u+1)\sqrt{2-u}$, so $dt = \dfrac{-b\,du}{\sqrt{2V_0/m}\,(u+1)\sqrt{2-u}}$. Substitute $w = \sqrt{2-u}$: $u = 2 - w^2$, $du = -2w\,dw$, $u + 1 = 3 - w^2$, with $w = 0$ at the start ($u = 2$) and $w \to \sqrt{3}$ at the barrier ($u \to -1$):

$$t(w) = \frac{2b}{\sqrt{2V_0/m}} \int_0^{w} \frac{dw'}{3 - w'^2}$$

With the antiderivative $\int \frac{dw}{3-w^2} = \frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3}+w}{\sqrt{3}-w}$ and Barrow's rule (the lower limit gives $\ln 1 = 0$):

$$t(w) = b\sqrt{\frac{m}{6V_0}}\;\ln\frac{\sqrt{3}+w}{\sqrt{3}-w} \xrightarrow[w\to\sqrt{3}]{} \infty$$

For instance, P passes $x = 0$ ($w = \sqrt{2}$) at $t = b\sqrt{m/(6V_0)}\,\ln\big(5 + 2\sqrt{6}\big) \approx 0.936\,b\sqrt{m/V_0}$, but it needs an infinite time to reach $x = -b$. The particle tends to the unstable equilibrium and never escapes: the statement is FALSE.

KEY POINTS

- FALSE.
- $V(2b) = 2V_0 = V(-b)$: the energy equals the top of the barrier. $F(2b) = -9V_0/b < 0$, so P does start to the left.
- $E - V = V_0(u+1)^2(2-u)$ with $u = x/b$: $|\dot{x}| \propto |x+b|$ near $x = -b$ and the time $t = b\sqrt{m/(6V_0)}\,\ln\dfrac{\sqrt{3}+w}{\sqrt{3}-w}$, $w = \sqrt{2 - x/b}$, diverges as $x \to -b$: P approaches the barrier top asymptotically and never crosses it.

Part (b) · 1 point

Released at $x = 0$

Statement: "If P is released from rest at $x = 0$, it oscillates about the stable equilibrium $x = b$ between $x = 0$ and $x = 2b$, symmetrically about $x = b$."

Step 1. Energy and initial force

$E = V(0) = 0$. The initial force is $F = -\frac{dV}{dx}\big|_0 = -\frac{3V_0}{b}(0-1) = \frac{3V_0}{b} > 0$: P moves to the right, into the well of the minimum $x = b$.

Step 2. Turning points

They solve $V(x) = E = 0$: $V_0(u^3 - 3u) = V_0\,u\,(u-\sqrt{3})(u+\sqrt{3}) = 0$, so $u \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Between $u = 0$ and $u = \sqrt{3}$ we have $V < 0 = E$ and the kinetic energy is positive; the motion is confined to $0 \le x \le \sqrt{3}\,b$. At both ends $dV/dx \neq 0$ ($-3V_0/b$ at $x = 0$, $+6V_0/b$ at $x = \sqrt{3}\,b$), so they are simple turning points reached in a finite time, and the motion is a periodic oscillation (Notes §4.5).

Step 3. Not symmetric, not up to $2b$

$x = 2b$ is never reached, since $V(2b) = 2V_0 > E$. The interval $[0, \sqrt{3}\,b]$ extends b to the left of the minimum and only $(\sqrt{3}-1)\,b \approx 0.732\,b$ to its right: the cubic well is steeper on the right. Symmetry about the minimum holds only for a parabolic well, i.e. for small oscillations, whose frequency follows from $V''(b) = \frac{1}{b^2}\frac{d^2V}{du^2}\big|_{u=1} = \frac{6V_0}{b^2}$: $\omega^2 = \dfrac{6V_0}{mb^2}$ (Notes §4.6). The statement is FALSE.

KEY POINTS

- FALSE.
- $E = V(0) = 0$; $V = V_0 u(u-\sqrt{3})(u+\sqrt{3})$: turning points $x = 0$ and $x = \sqrt{3}\,b \approx 1.732\,b$ (not $2b$, where $V = 2V_0 > E$).
- $x = b$ is stable ($V'' > 0$), small-oscillation frequency $\omega^2 = 6V_0/(mb^2)$; the motion is periodic but the distances to the turning points are b and $0.732\,b$.

Interpretation and checks

- **Dimensional check:** V_0/b is a force (N); $b\sqrt{m/V_0}$ is $\mathrm{m}\sqrt{\mathrm{kg/J}} = \mathrm{s}$; $\omega^2 = 6V_0/(mb^2)$ is s^{-2}.
- **Limits:** with E slightly below $2V_0$, P would turn back just before $x = -b$ after a long but finite time; slightly above, it would cross the barrier and run to $x \to -\infty$ ($V \to -\infty$, unbounded speed). $E = 2V_0$ is the separatrix between both behaviours.
- **Common mistakes in the exam:** (1) assuming that "enough energy to reach the top" means crossing it; (2) placing the turning points symmetrically about the minimum; (3) forgetting to check the sign of the initial force to know where the particle goes.